

Chemical Mixing Guide

Lime · PACI Coagulant · Flocculant · SUMAS Water Treatment, Kamloops BC
Created by Oliver Peters · June 2026

1. Lime

1 bag (50 lbs) per 200L of water



TEXLIME High Calcium Hydrated Lime — confirmed on-site bag

1. Add 200L of clean water to the mixing tote.
2. Add 1 bag (50 lbs) of lime.
3. Turn on the air mixer (open the red valve) — let it blend.
4. **Keep the mixer running the whole time you're pumping it into the system.** If the blade stops, the lime settles and hardens fast, and the dose won't work.

Critical: this is not "mix once and walk away." The blade has to stay on for as long as you're drawing lime out — mixing and pumping happen at the same time.

2. PACI Coagulant

1 bag (25kg dry powder) per 100L of clean water



Poly Aluminum Chloride, 25kg bag — confirmed on-site bag

1. Add 100L of clean water to the mixing tote (not the water being treated).
2. Add 1 bag (25kg) of dry PACI powder.
3. Mix until dissolved.
4. **Let it sit 3 hours before using** — don't dose straight from a freshly mixed batch.
5. Meter the resulting solution into the process with the dosing pump.

Plan ahead: mix a new batch before you run out mid-shift — the 3-hour rest means you can't make more on demand.

3. Flocculant (Flopham Dry Polymer)

100g per 200L of water (~500 mg/L) — half a scoop from the white pail



Flopham water-soluble polymer — confirmed product label

1. Add 200L of clean water to the mixing tote.
2. Turn the mixer on first (open the red valve — go gradual for this one, see Section 4 below).
3. Sprinkle the Flopham powder in slowly while the mixer keeps running — never dump it in all at once.
4. Keep mixing until it looks clear and uniform — about 1 hour.
5. Meter the resulting solution into the process alongside PACl. **Rule: flocculant dose must always be equal to or more than the coagulant dose — never less.**

Never dump dry polymer straight into still water — it forms lumps that won't dissolve. Always pour it in slowly while the mixer is running.

 Updated: this replaces an earlier 1kg/1000L, 24-hour-rest ratio logged here for a different product/account. The Flopham dry polymer actually in routine use at the C Can is mixed at 100g/200L and only needs about 1 hour before use — see Section 5.5 of the full manual.

4. The Mixer (Air-Driven Tote Mixer)



Mixer clamped on top of the tote



Close-up: red air valve, shaft & propeller

1. Connect the air hose to the compressed air supply (quick-connect fitting).
2. Open the **red-handled valve** — air spins the motor, which turns the propeller inside the tote.
3. Open it gradually for flocculant — full-blast air can tear the polymer apart instead of dissolving it. Lime and PACl can run full open.
4. Close the red valve to stop.

Don't confuse it with the nearby cross-handled valve — that one's the tote's drain/outlet, not the mixer.

Full detail and safety info: Section 5 (Chemical Dosing) and Section 6.16 (Equipment Guide) of the full manual.